

Microtubules

An Overview

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Summary

Microtubules are found in all eukaryotes and are built from $\alpha\beta$ -tubulin heterodimers. The α -tubulins and β -tubulins are among the most highly conserved eukaryotic proteins. Other members of the tubulin family have come to light recently and, like γ -tubulin, appear to play roles in microtubule nucleation and assembly. Microtubule assembly is accompanied by hydrolysis of GTP associated with β -tubulin so that microtubules consist principally of “GDP-tubulin” stabilized by a short “GTP cap.” Microtubules are polar, cylindrical structures some 25 nm in diameter. Protofilaments made from tubulin heterodimers run lengthwise along the microtubule wall with the β -tubulin subunit at the microtubule plus end. The crystallographic structures of tubulins are essential to understand in detail microtubule architecture and interactions with stabilizing and destabilizing drugs and proteins.

Key Words: Tubulin family; tubulin structures; microtubule assembly; microtubule structure.

1. Introduction

Microtubules are key actors in the cytoskeleton of eukaryotic cells where they play important roles in organizing the spatial distribution of organelles throughout interphase and of chromosomes during cell division. They can be extremely stable as in cilia and flagella or very dynamic as in the mitotic spindle. Microtubules are approx 25-nm diameter hollow tubes with walls made from tubulin heterodimers ($\alpha\beta$ -tubulin) stacked head-to-tail to form “protofilaments” that run lengthwise along the wall of the tube conferring the structural and a dynamic polarity that is an essential feature of microtubules. When microtubules assemble, one end (called the plus end) grows more quickly than the other. In the cell, microtubules usually grow outwards from the microtubule-organizing center (MTOC) towards the cell membrane with the plus

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end leading. They often remain attached to the MTOC by their minus end and the resulting microtubule networks engage in intracellular transport, in cell division and, in the form of rings of doublets, in ciliary and flagellar motility. They interact with many ligands including the dynein and kinesin motor proteins for which they provide directional pathways. Molecules isolated from plants can strongly influence microtubule assembly and stability. Such drugs have important fundamental and medical applications.

2. Tubulin Family

2.1. α -Tubulin and β -Tubulin Go Together

The α -tubulin and β -tubulin monomers have similar masses of about 55 kDa as measured on sodium dodecyl sulfate-polyacrylamide gels and they both have about 450 amino acid residues. The monomers interact noncovalently to form the very stable tubulin heterodimer, which is the functional form of the protein. Tubulin, actin, and elongation factor (EF)-1 α are among the most highly conserved eukaryotic proteins and their amino acid sequences have been used to construct phylogenetic trees of eukaryotic organisms that can be compared to trees obtained by analysis of small ribosomal subunit RNA (*1*). Because of the expression of several tubulin genes and topf posttranslational modifications, several isoforms of α - and β -tubulin often coexist in eukaryotic cells. These evolutionarily conserved tubulin isotypes are distinguished by a characteristic stretch of some 15 or more amino acid residues, rich in acidic amino acids, at their COOH-terminus and also to some extent by their NH₂-terminal region. Different α - and β -tubulin isotypes with highly conserved sequences are often expressed independently, and each isotype tends to have similar gene expression patterns within and between species. Different organisms possess various numbers of evolutionarily conserved isotypes, some unicellular organisms have a unique $\alpha\beta$ -tubulin, others, like the vertebrates, can have a large number of tubulin genes and many pseudogenes (*2,3*). The degree of sequence conservation within the α - and β -tubulin subfamilies is around 60% and there is typically around 40% similarity between the two types of tubulin. There are distinct sequence differences in a few specific regions of α - and β -tubulin.

Tubulin undergoes a number of posttranslational modifications, including for example, acetylation of a specific lysine near the NH₂-terminus of α -tubulin by tubulin acetyltransferase, and removal of the C-terminal tyrosine residue of α -tubulin by a tubulin specific carboxypeptidase (tubulin detyrosinase) that acts preferentially on polymerized tubulin. Another enzyme, tubulin-tyrosine ligase can reattach tyrosine to the C-terminus of α -tubulins. Up to seven glutamate residues (polyglutamylation) can be added to a glutamate residue near the C-terminus of both α - and β -tubulin. Other posttranslational modifications include phosphorylation and polyglycylation.

There are at least three alternative explanations for the existence of multiple tubulin isotypes. Each has a measure of experimental support, implying that the isotypes probably have multiple biological functions (2,4,5).

Explanation 1: the isotypes result from a slow sequence drift of duplicated genes and have no functional significance. In this case, isotypes should be functionally interchangeable and there are several reports appearing to support this.

Explanation 2: the existence of isotypes could improve an organism's capacity to respond to environmental challenges. For example, the temperature dependent transcription of some of the nine β -tubulin isotypes in the plant *Arabidopsis thaliana* could favor adaptation to extreme conditions.

Explanation 3: individual tubulin genes code for proteins participating in a set of distinct $\alpha\beta$ heterodimers each with a specific microtubule function. This is known as the multitubulin hypothesis. Among several clear examples of different $\alpha\beta$ -tubulins influencing microtubule structure and function, the nematode *Caenorhabditis elegans* is particularly striking. Somatic cells in this nematode have microtubules with 11 protofilaments but, in the six touch cells (mechanosensory neurons), the highly expressed tubulins mec-12 (α -tubulin) and mec-7 (β -tubulin) participate, as a cell specific tubulin heterodimer, in the formation of 15 protofilament microtubules. These have been shown by mutational studies to be essential for touch sensitivity (5–7).

2.2. Other Tubulins Like Microtubule Minus Ends

It was a surprise when, in 1989, a new tubulin with only about 30% identity to α - and β -tubulins was discovered in the fungus *Aspergillus nidulans* (8). This protein forms a distinct subfamily, γ -tubulin, that appears to exist in all eukaryotes where it is involved in microtubule nucleation and stabilization. It is located in MTOC (3,9). In many cells, γ -tubulin is found associated with a number of other proteins in a 2.2-MDa tubulin ring complex (g-TuRC). A smaller tubulin complex (g-TuSC) also exists. A number of in vitro experiments have confirmed the role of γ -TuRCs in microtubule nucleation from centrosomes and spindle pole bodies. Both γ -tubulin and γ -TuRCs have been shown to bind to microtubule minus ends. Although γ -tubulin in complex with pericentrin (10,11) is involved in microtubule nucleation within MTOCs a detailed picture of microtubule nucleation and early growth remains a challenge.

Four other tubulins have now been identified (3,5) and they all appear to localize to the centriole or basal body regions or to affect the functions of these "organelles." A δ -tubulin mutant leads to defective flagella assembly from basal bodies in the green alga *Chlamydomonas*. ϵ -Tubulin locates to the region around the centriole in a cell cycle dependent manner but is not directly involved in microtubule nucleation. η -Tubulin may tether γ -tubulin to basal bodies and a mutation of this gene in *Paramecium* leads to basal body duplication defects.

Finally, ζ -tubulin is found in the basal body region in trypanosomes and in the centriolar region in some animal cells. Significantly, these tubulins all appear to be involved in the duplication of centrioles and basal bodies (3).

2.3. Prokaryotes Have Tubulin “Ancestors”

FtsZ (filamenting temperature sensitive strain) is a 40-kDa protein that forms filamentary structures in bacteria. It is involved in cell division in many bacteria by forming, together with other proteins, a contractile ring at the site of septation (12). FtsZ is ubiquitous in eubacteria, and is also found in some archaeobacteria, in chloroplasts and in some mitochondria. It binds and hydrolyses GTP. It has limited sequence similarity to tubulin (typically around 33% identical/strongly similar residues in amino acid sequence alignments) with a higher degree of similarity in the approx 250 N-terminal residues. The nucleotide binding motif GGGTGTG is very similar to the tubulin signature motif [SAG]GGTG[SA]G (13). Typical FtsZ sequences are around 360 residues in length. The crystal structures of FtsZ and tubulin are very similar and these proteins appear to form a distinct GTPase family (14–16).

Two genes coding for tubulin-like proteins, BtubA and BtubB (bacterial tubulin a and b, respectively) have been reported in the bacterial genus *Prostheco bacter*. These prokaryotic tubulins have about 35% sequence identity to their eukaryotic counterparts and only around 10% identity to bacterial FtsZ (17,18). BtubA and BtubB have been expressed in *Escherichia coli*, allowing their assembly to be studied in vitro. When mixed together in the presence of GTP, they form, in a 1:1 ratio, long protofilament bundles, four to seven protofilaments wide. BtubA and BtubB genes, but not FtsZ, are present in four species of the *Prostheco bacter* genus. The very strong structural similarity to eukaryotic tubulins (19) supports horizontal gene transfer from a eukaryote.

3. Microtubule Assembly and Disassembly

The purification of tubulin in large quantities starting from pig or bovine brain where it makes up some 20% of the protein content was essential to allow widespread in vitro studies of many aspects of microtubule behavior and structure, for an example see ref. 20. Both α -tubulin and β -tubulin bind guanosine triphosphate but GTP is attached permanently at the N-site (nonexchangeable) in α -tubulin and only the β -subunit is active as a GTPase with exchangeable nucleotide attached to the E-site. Tubulin assembly requires GTP, magnesium ions and a temperature higher than approx 30°C. GTP hydrolysis occurs as microtubules assemble. The overall mass of assembling microtubules can be followed by measuring the solution turbidity in a spectrophotometer. Microtubules disassemble at low temperatures and cycling between soluble tubulin at 4°C and microtubules at 37°C is an important part of tubulin purification protocols.

Because GTP hydrolysis accompanies their growth, microtubules are mostly made up of “GDP-tubulin” with the growing end capped by GTP tubulin, probably limited to the endmost tubulin dimer (21,22). Microtubules are also formed in the presence of the slowly hydrolyzable GTP analogue GMPCPP, showing that GTP hydrolysis is not in itself essential for microtubule growth (23). “GMPCPP” microtubules are more stable than “GDP” microtubules, suggesting that GDP-tubulin is required for microtubule disassembly. Although constrained to be straight within the microtubule wall, GDP-tubulin has long been thought to have a natural curved conformation as witnessed by the ring-like structures produced when microtubules lose their GTP-cap and disassemble. Electron cryomicroscopy shows how, during disassembly, protofilaments peel off microtubule extremities to form circular oligomers some 34 nm in diameter (24). These GDP-tubulin oligomers can then recycle into assembly competent tubulin dimers by exchanging GTP for GDP (25).

The growth of individual microtubules can be followed using various light microscopy techniques (digital interference contrast microscopy [DIC], fluorescence microscopy, dark field microscopy). The first direct observations of *in vitro* microtubule assembly in 1986 revealed that individual microtubules both grow and shrink actively (26). Plus ends grow about 10 times more quickly than minus ends. Microtubule growth at both ends is interrupted at random times by “catastrophic” shrinkage and then by “rescue” initiating a new phase of rapid growth. This behavior is called dynamic instability and was previously proposed by Mitchison and Kirchner to explain the behavior of microtubules during mitosis (27). Dynamic instability is now accepted as an important aspect of microtubule behavior both in interphase and in the establishment of interactions between the microtubule spindle and chromosomes during mitosis.

In interphase cells, microtubules are usually nucleated in the region surrounding the centrosome (MTOC) that is rich in ring-like structures containing γ -tubulin. During cell division, microtubules nucleate at the two spindle poles, where γ -tubulin complexes are also located, then grow and shrink dynamically. They ultimately attach to kinetochores on chromatid pairs and align chromosomes to the midplane at metaphase. Interestingly, kinetochores bind preferentially to GTP- rather than GDP-microtubules *in vitro*, and attach to the plus-end in preference to other positions on the sides or at the minus end of microtubules (28).

Microtubule networks, and the polar distribution of microtubules in the mitotic spindle, provide pathways for dynein and kinesin allowing these ATP dependent microtubule motors to participate in cell division and to engage in intracellular transport and organization. Both motor proteins can detect the underlying structural polarity of microtubules and move in specific directions with unerring accuracy (29).

4. Microtubule Organization and Structure

The quest for information on microtubule structure has extended from the 1960s until the present. Initial successes were obtained by electron microscopy of (1) thin sections of glutaraldehyde/tannic acid fixed cells from various organisms and, (2) of negatively stained axonemal microtubules. These observations helped to establish the complex organization of cilia and flagella based on nine microtubule doublets encircling a central pair of microtubules. As thin section fixation and staining methods improved it became possible to directly count thirteen “protofilaments” in cross sections of the cylindrical, 25-nm diameter, microtubule wall. **Images of individual microtubules obtained using negative stain showed the protofilaments are aligned lengthwise along the microtubule and associate laterally with an approx 0.9 nm shift from one protofilament to the next.** As a result, tubulin subunits in neighboring protofilaments describe a 12-nm pitch, left-handed, helical pathway around the microtubule. Subsequently, microtubules with different numbers of protofilaments have been found in a variety of organisms and specialized cells. Genetic methods have shown β -tubulin is particularly important for determining microtubule architecture (6,7,30).

Microtubules assembled from pure tubulin in vitro have a wide range of protofilament numbers, typically in the range 10 to 16 (31,32). The moiré pattern contrast of well preserved microtubules observed in their fully hydrated state by electron cryomicroscopy shows that the microtubule architecture allows this wide range of protofilament numbers to be readily accommodated by a global rotation of the surface lattice. **Protofilaments run parallel to the microtubule axis only for 13 protofilament microtubules, in all other cases the protofilaments describe long-pitch helices.**

Figure 1 represents a three-dimensional reconstruction of microtubules at about 14 Å resolution obtained by electron cryomicroscopy and computerized three-dimensional reconstruction methods (32). The outer surface shows parallel, compact protofilaments separated by deep grooves. The approx 40 Å (4 nm) periodicity visible along each protofilament corresponds to the spacing between the α - and β -tubulin subunits that are indistinguishable at the resolution of the reconstruction. The main contacts between protofilaments are at the bottom of the grooves that are pierced by holes about 15 Å by 20 Å in size giving direct access to the microtubule lumen. The outermost surface of the protofilaments appears quite smooth and the inner surface significantly rougher.

Much effort has been made to relate the tubulin dimer orientation to microtubule polarity and **it is now established that β -tubulin caps the microtubule plus end and α -tubulin the minus end** (33–36). Growth at the microtubule plus end is achieved by the interaction of the α -subunit of an incoming tubulin dimer

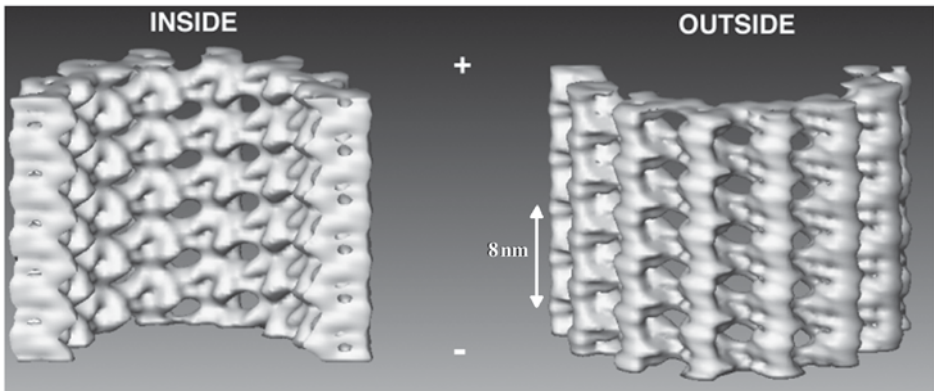


Fig. 1. Microtubule structure obtained by electron cryomicroscopy and three-dimensional image reconstruction (32). The surface representation is been cut length-wise to show the both the inner and outer surfaces. Protofilaments run vertically, with the microtubule plus end at the top. Contacts between the protofilaments are seen to establish across the deep grooves between the protofilaments. The inner microtubule surface is significantly “rougher” than the outside.

with an *in situ* β -subunit thereby activating hydrolysis of GTP at the E-site in β -tubulin. The α -tubulin capped minus end will behave differently because GTP in α -tubulin is buried within the $\alpha\beta$ interface and is neither exchangeable nor hydrolyzable. The organization of the microtubule surface lattice is directly visible in three-dimensional reconstructions using images obtained by electron cryomicroscopy of microtubules “decorated” with kinesin motor domains. Identical tubulin subunits are in contact from protofilament to protofilament around the microtubule wall. This organization is known as the B lattice (37,38). Importantly, this implies that standard 13 protofilament microtubules have a lattice mismatch, or “seam,” between a single pair of protofilaments (Fig. 2) (38,39). This is a direct consequence of the microtubule assembly mechanism involving endwise addition of tubulin dimers onto a sheet-like lattice at the microtubule extremity. This sheet rolls up into a cylinder as the microtubule grows (40). Consequently, microtubule growth is not a helical process and microtubules are cylindrical, rather than helical, structures. The seam may turn out to be important for microtubule disassembly by providing a “weak link” facilitating the peeling apart of protofilaments.

5. Tubulin at Atomic Resolution

An atomic resolution structural model of the tubulin heterodimer was obtained in 1998 by electron crystallography of taxol stabilized tubulin sheets obtained in

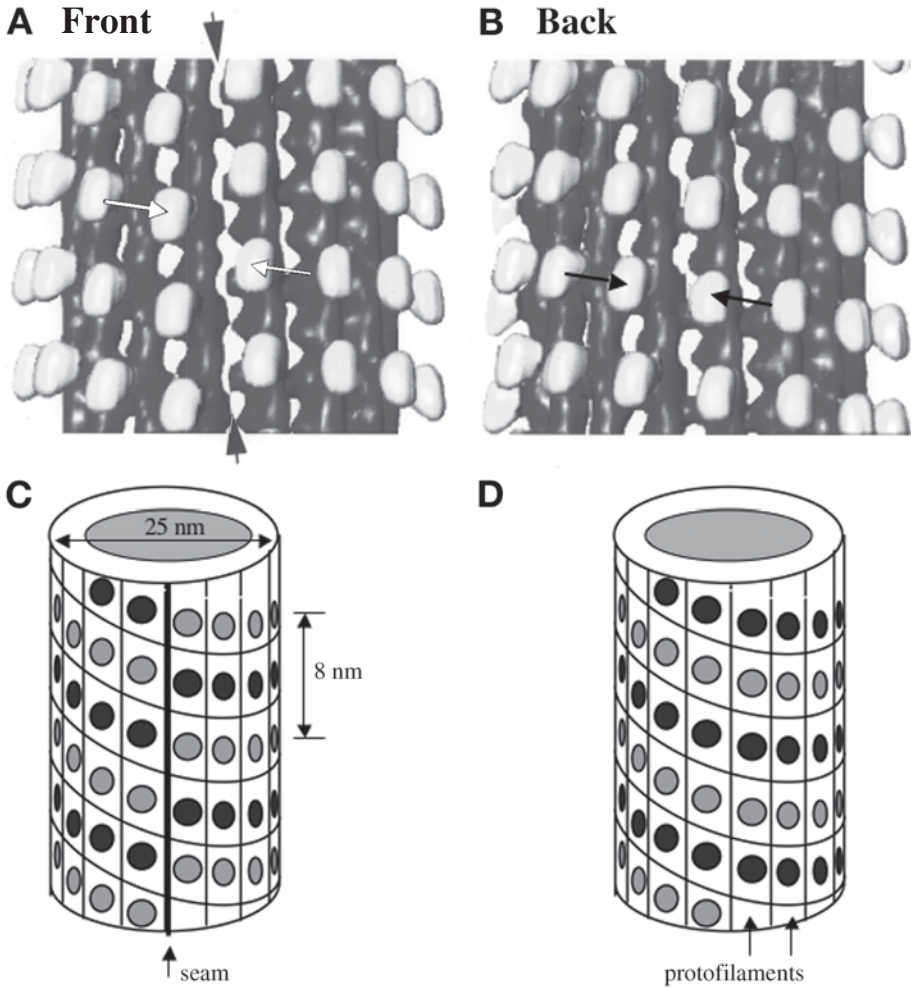


Fig. 2. Microtubule organization. The surface lattice organization can be visualized in three-dimension image reconstructions of microtubules specifically decorated with monomeric kinesin motor domains. Using images obtained by electron cryomicroscopy and tomographic reconstruction methods, similar tubulin subunits are highlighted by kinesin monomers and follow a helical path around the microtubule. This is interrupted by a seam (arrows), compare the views of both sides of the reconstruction (A,B). (C,D) Drawings of front and back views of a 13 protofilament microtubule, α -subunits black and β -subunits gray. Protofilaments run vertically as stacks of tubulin heterodimers. Similar subunits follow a helical path around the cylindrical microtubule wall and after a full turn the subunits switch from one type to the other across the so-called “seam.”

the presence of zinc ions (41). The protofilaments in these sheets have the same longitudinal organization as in microtubules, giving a putative model of the inter- and intradimer contacts along the protofilaments. However, lateral contacts between protofilaments are different in the two lattices because neighboring protofilaments are aligned antiparallel in “zinc” sheets and parallel in microtubules.

The crystallographic structures of the α - and β -subunits are practically identical (Fig. 3A). Both have a compact structure with three functional domains. The N-terminal domain forms a Rossmann fold with six α -helices and a six-stranded parallel β -sheet with seven loops involved in nucleotide binding. The α -helix H7 connects the N-terminal and the intermediate domain that runs from residues 206–381 (Fig. 3B). This domain consists of a four-stranded, mainly parallel, β -sheet flanked by two α -helices on one side and one on the other. In β -tubulin this domain includes the taxol-binding pocket. The C-terminal domain has two long, almost parallel, α -helices (H11 and H12) running along the full length of each subunit. The C-terminal residues (10 in α -tubulin and 18 in β -tubulin) are disordered and are not visible in the structures.

The α -tubulin subunit has GTP in the nucleotide binding site (N-site) whereas the E-site in β -tubulin contains GDP. Taxol is also visible in the structure of β -tubulin. Significantly, the nucleotide is positioned at the interface between monomers along the protofilaments. The α -tubulin N-site is hidden within the intra dimer interface both in the free heterodimer and in assembled microtubules. The β -tubulin GTP binding site is partially exposed in the heterodimer explaining how nucleotide exchange is possible in solution but impossible when this site is buried within microtubule protofilaments.

Recently the crystallographic structure of tubulin complexed with the α -helical stathmin-like domain (SLD) of the neural protein RB3 has been determined by X-ray diffraction (42). In this structure, the helical segment of RB3-SLD binds two $\alpha\beta$ -tubulin heterodimers in a curved configuration and the NH₂-terminal region that is conserved in the stathmin family forms a β -hairpin capping the endmost α -tubulin subunit. Interestingly, the tubulin structures have a distinct “curved” conformation compared to the “straight” conformation of the zinc sheet structures. Notably, the intermediate domains are rotated by about 10° in the curved conformation and there are local differences in loops involved in contacts along and between protofilaments.

Crystals of the tubulin-RB3-SLD complex are proving to be a very powerful tool for examining the interaction of tubulin with microtubule destabilizing drugs (Fig. 4). The colchicine binding site is found to be at the interface between the α - and β - subunits of the same tubulin heterodimer (43). Colchicine is mostly buried in the β -tubulin intermediate domain between β -strands S8, S9, loop T7

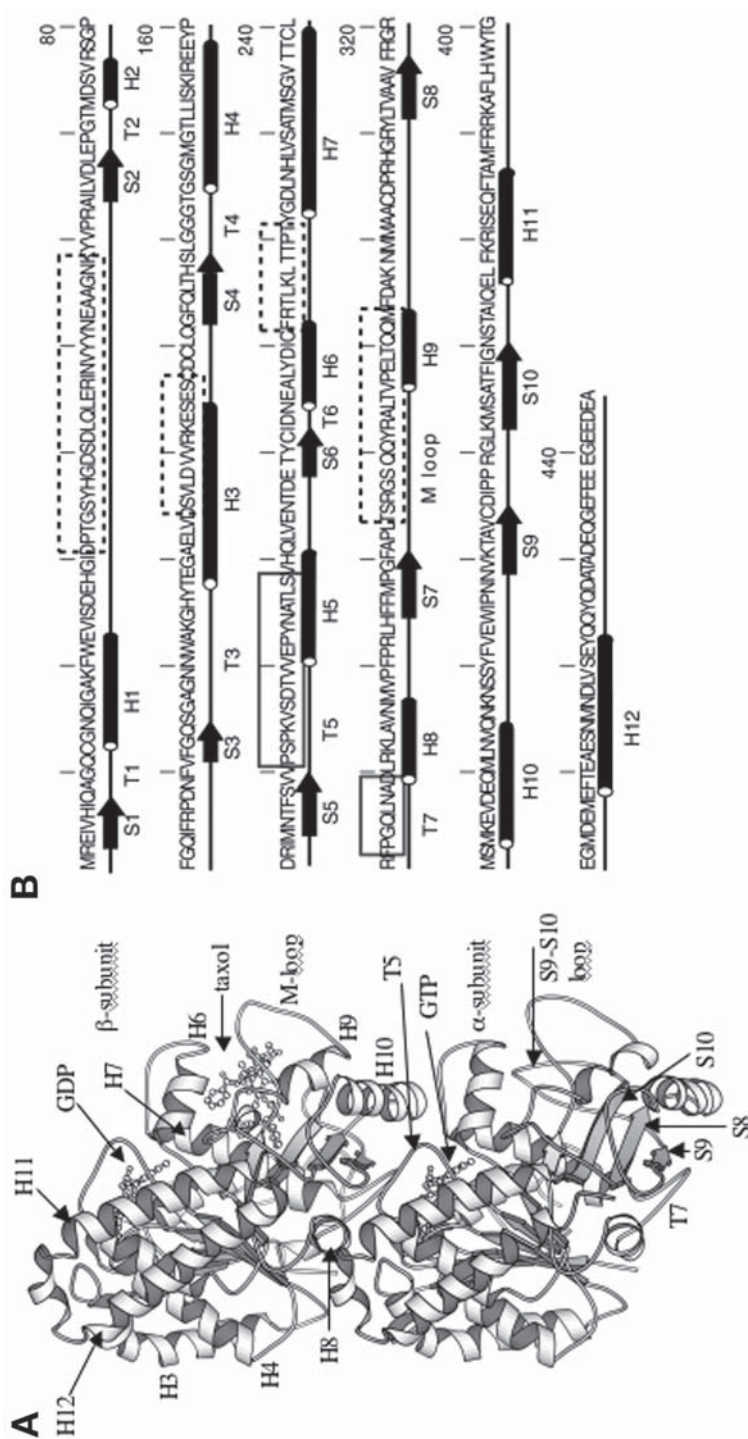


Fig. 3. (A) Structure of the tubulin heterodimer obtained by electron crystallography (41). Key structural features are indicated, β -strands S, α -helices H, and turns T. Note that GTP binds near the top of the subunits where it is exposed in the β -subunit whereas the α -tubulin nucleotide is enclosed between the two subunits. There is a taxol binding pocket in β -tubulin whereas this region in α -tubulin is occupied by longer loop between S9 and S10 (B) Sequence of pig brain β -tubulin with corresponding structural features. The boxed regions are important for lateral (dashed lines) and longitudinal contacts (full lines), respectively.

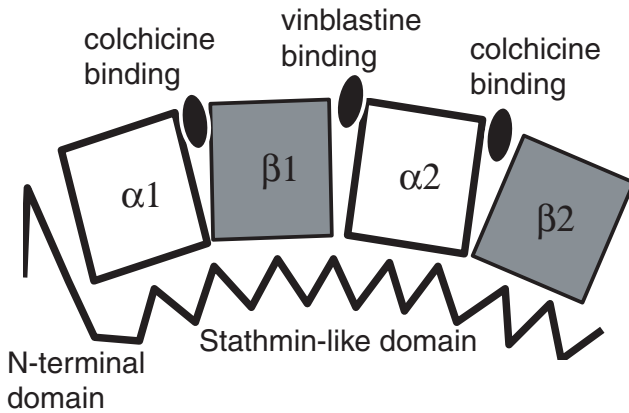


Fig. 4. Schematic representation of the curved configuration of two tubulin dimers, $\alpha 1\beta 1$ and $\alpha 2\beta 2$, interacting with a stathmin-like helix based on the structure of a tubulin–RB3–SDL complex determined by X-ray crystallography (42). The curved tubulin conformation may prevent lateral contacts between protofilaments thus explaining tubulin sequestration by the stathmin-related proteins. Colchicine has been shown to bind between subunits of the same dimer whereas vinblastine binds between dimers (42–44).

and α -helices H7, H8. It interacts with loop T5 of the α -tubulin subunit and locks-in the curved tubulin conformation. Podophyllotoxin was shown to bind to the same site. On the other hand, the vinca alkaloid, vinblastine has been known for some time to bind to a different site and this has now been localized in the tubulin–colchicine–RB3–SLD complex by soaking crystals in vinblastine solutions before flash freezing (44). The binding site lies between two tubulin heterodimers and includes tubulin residues located toward the inner lumen of microtubules and involved in longitudinal contacts along microtubules. In solution, vinblastine is known to induce the formation of spiral-like tubulin aggregates. This is also the case for a tubulin–RB3–SLD complex in which 24 amino acid residues have been removed from the NH_2 -terminus of the RB3–SLD.

The crystal structure of a human γ -tubulin GTP- γ S complex was obtained recently at 2.7 Å resolution (45). This is the first structure of a monomeric tubulin and has the highest resolution obtained to date. Although γ -tubulin is in the GTP state (GTP- γ S is a nonhydrolysable GTP analogue) it has a curved configuration similar to that observed in the $\alpha\beta$ -tubulin–RB3–SLD complex. One of the crystal packing interactions resembles the lateral interactions between protofilaments in microtubules showing that monomeric γ -tubulin

in the curved configuration might self-associate by forming stable lateral interactions. This may allow the γ -tubulin template in the γ -TuRC to nucleate microtubules by interacting longitudinally with $\alpha\beta$ -tubulin whilst enhancing lateral interactions between the tubulin heterodimers.

6. Microtubules at Pseudo-Atomic Resolution: Interactions With Drugs and Proteins

Although a microtubule structure has now been solved at 8 Å resolution (46), the best way to obtain a microtubule model at pseudo-atomic resolution is to dock the tubulin dimer structure obtained by crystallography into a three-dimensional microtubule reconstruction obtained by electron cryomicroscopy (32,47). The most important feature revealed by such models concerns the lateral interactions between tubulin subunits in adjacent protofilaments (Fig. 5). These interactions take place mainly in the regions at the bottom of the grooves between protofilaments where the three-dimensional image reconstructions show significant density. The M-loop situated between β -strand S7 and α -helix H9 in one tubulin subunit interacts with H3 and loops H1-S2, H2-S3 in the tubulin subunit in the neighboring protofilament. In β -tubulin, this loop is influenced by the antimetabolic drug, taxol, that binds in a nearby pocket, occupied in α -tubulin by an eight-residue insert in the loop between S9 and S10 (41,47). The stabilization of microtubules by taxol blocks cell division leading to the use of this drug for chemotherapy of certain forms of cancer (48).

Microtubules assembled in the presence of taxol have fewer protofilaments than a drug-free control assembly, suggesting that taxol influences the M loop configuration leading to both microtubule stability and a reduced number of protofilaments. Note that taxol resistant human ovarian cancer cells have β -tubulins with mutations near the beginning of the M-loop and in the S9-S10 loop (49).

Other microtubule stabilizing agents including the epothilones, sarcodictyin A, eleutherobin also influence microtubule architecture. Sarcodictyin A, epothilone B, and eleutherobin all modify the average numbers of protofilaments in vitro assembled microtubules so apparently, like taxol, they also effect lateral interactions between tubulin subunits in adjacent protofilaments. Taxotere increases the average protofilament number, indicating that this drug must have different interactions within the taxol-binding region of β tubulin. Interestingly, GMPCPP, a nonhydrolysable GTP analogue, is unique in producing microtubules with a well-defined number of protofilaments. Typically 96% of the microtubule population has 14 protofilaments. The implication that the nucleotide influences lateral interactions between protofilaments is, at first site, surprising because the exchange-

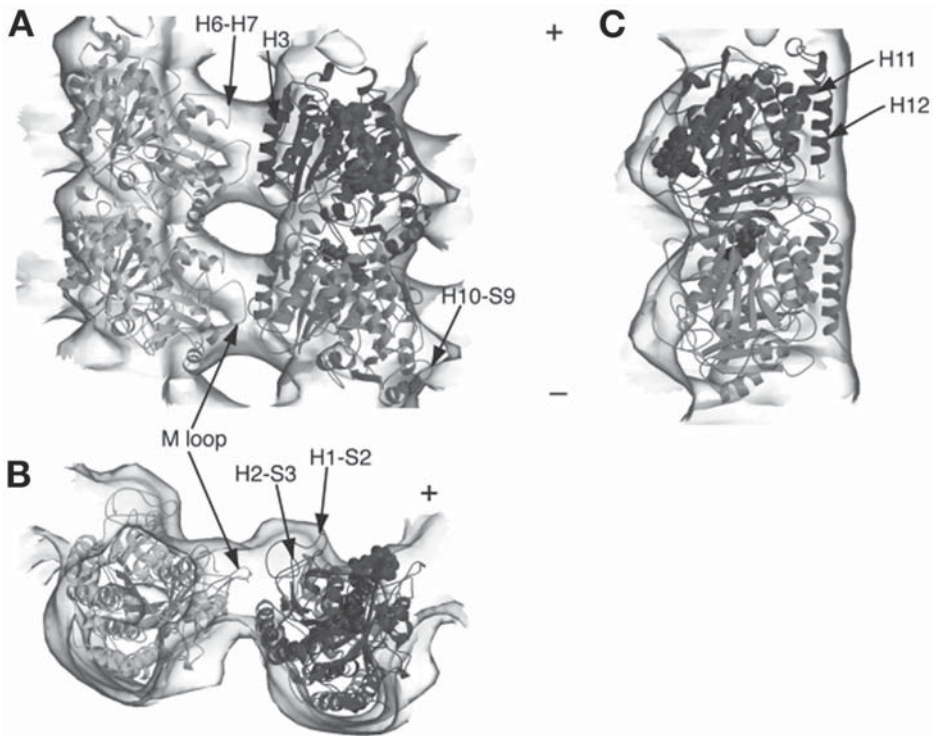


Fig. 5. The tubulin crystal structure docked into the microtubule map, gray isodensity surface. The α - and β -tubulin subunits are shown, respectively, as lighter and darker gray ribbons. (A) Part of two protofilament neighbors viewed from the outer surface of the microtubule, with the microtubule plus end at the top. (B) Protofilaments seen end-on, looking from the microtubule plus end. (C) Side view of a protofilament as seen across the thickness of the microtubule wall. Helices H11, H12 run along the smooth outer surface on the right.

able site on the β subunit lies at the longitudinal interface between tubulin dimers along the protofilaments. However, several regions directly involved with nucleotide binding could possibly effect the position of helices or loops involved in lateral interactions (32,41,47).

In addition to the data obtained by crystallography many other results support the idea that GTP-tubulin is straight and GDP-tubulin is bent as in the curved protofilaments that peel off the plus end of microtubules as microtubules disassemble (24,25). Interestingly, the existence of two tubulin conformations in microtubules was detected in an investigation showing the tubulin dimer spacing along microtubule protofilaments is slightly greater for tubulin-

GMPCPP (GTP configuration) than for tubulin-GDP (50). Because microtubules normally contain GDP-tubulin in the curved configuration it is the lateral interactions within the microtubule wall that hold GDP-tubulin in place until released during disassembly. The curved conformation observed for tubulin dimers complexed with the α -helical RB3 stathmin-like domain can be fitted to protofilaments at the microtubule plus end by computer modelling (32,42). The helical stathmin-like domain is found to be offset to one side of a protofilament provoking a curvature that prevents the lateral contacts between the M loop and the neighboring protofilament perhaps accounting for tubulin sequestration by proteins in the stathmin family.

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